

ENUNCIADO — PROBLEMA 1

Numerical Analysis Preliminary Examination questions
January 2013

Instructions: NO CALCULATORS are allowed. This examination contains **seven** problems, each worth 20 points. Do any **five** of the seven problems. Show your work. Clearly indicate which **five** are to be graded.

PLEASE GRADE PROBLEMS: 1 2 3 4 5 6 7

1. Consider the fixed point iteration method $x_{k+1} = g(x_k)$, $k = 0, 1, \dots$ for solving the nonlinear equation $f(x) = 0$. Consider choosing an iteration function of the form

$$g(x) = x - af(x) - b(f(x))^2$$

where a, b are parameters. Let z be a root of the $f(x)$. Show that $a = \frac{1}{f'(z)}$ and $b = -\frac{1}{2} \frac{f''(z)}{(f'(z))^3}$ for the iteration method to be of third order.

2. Consider the quadrature formula,

$$\int_a^b f(x) dx = h \left\{ af(0) + bf\left(\frac{h}{3}\right) + cf(h) \right\}.$$

- (a) Determine a, b and c such that the quadrature formula is exact for polynomials of as high order as possible.
- (b) If the truncation error of the formula is given by $\frac{C}{3!}f'''(\xi)$ where $0 < \xi < h$, then show that $C = -\frac{h^4}{36}$.

3. Consider approximating the function $f(x) = \sqrt{1+x}$ in the interval $[0, 1]$ at equally spaced nodes $x_j = jh, j = 0, 1, \dots$. Let the quadratic Lagrange interpolating polynomial to $f(x)$ at x_{j-1}, x_j, x_{j+1} be $P_2(x)$. Show that the error in this approximation satisfies:

$$|f(x) - P_2(x)| \leq \frac{h^3}{24\sqrt{3}}$$

4. Each matrix A and B shown below has determinant equal to ϵ^2 when computed using exact arithmetic,

$$A = \begin{bmatrix} \epsilon & 0 \\ 0 & \epsilon \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 1+\epsilon \\ 1-\epsilon & 1 \end{bmatrix},$$

where ϵ is a small positive constant.

Suppose that instead of using exact arithmetic you use a computer with floating point arithmetic to compute the determinant of A and B as ϵ gets smaller and smaller. Assume that your computer has machine precision $\epsilon_{mach} \approx 10^{-16}$, an underflow level, $UFL \approx 10^{-308}$ and an overflow level, $OFL \approx 10^{308}$. At what approximate value of ϵ would you expect each matrix to be numerically singular? Note that your floating point calculation should start with the matrix – not the exact formula ϵ^2 – to compute each determinant. Explain your reasoning for each matrix.

SOLUCIÓN — PROBLEMA 1

Problema 1 — Iteración de punto fijo de tercer orden

$g(x) = x - af(x) - b(f(x))^2$. Sea z la raíz de f y $e_k = x_k - z$.

Para que la convergencia sea de **orden 3** necesitamos $g'(z) = 0$ y $g''(z) = 0$.

Condición $g'(z) = 0$:

$$g'(x) = 1 - af'(x) - 2bf(x)f'(x).$$

En z , $f(z) = 0$: $g'(z) = 1 - af'(z) = 0 \Rightarrow a = \frac{1}{f'(z)}$.

Condición $g''(z) = 0$:

$$g''(x) = -af''(x) - 2b[(f'(x))^2 + f(x)f''(x)].$$

En z : $g''(z) = -af''(z) - 2b(f'(z))^2 = 0$. Sustituyendo $a = 1/f'(z)$:

$$-\frac{f''(z)}{f'(z)} - 2b(f'(z))^2 = 0 \Rightarrow b = -\frac{f''(z)}{2(f'(z))^3} = \boxed{-\frac{1}{2} \frac{f''(z)}{(f'(z))^3}}.$$

Con $g'(z) = g''(z) = 0$ y (generalmente) $g'''(z) \neq 0$, por Taylor: $e_{k+1} = \frac{1}{6}g'''(z)e_k^3 + O(e_k^4)$: **convergencia de orden 3.** ■

ENUNCIADO — PROBLEMA 2

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SOLUCIÓN — PROBLEMA 2

Problema 2 — Cuadratura de 3 puntos: coeficientes y error

(a) $\int_0^h f \, dx = h\{af(0) + bf(h/3) + cf(h)\}$ exacta para $f = 1, x, x^2$:

- $f = 1$: $h = h(a + b + c) \Rightarrow a + b + c = 1$.
- $f = x$: $h^2/2 = h^2(b/3 + c) \Rightarrow b/3 + c = 1/2$.
- $f = x^2$: $h^3/3 = h^3(b/9 + c) \Rightarrow b/9 + c = 1/3$.

Restando la 3ª de la 2ª: $(2b/9) = 1/6 \Rightarrow b = 3/4$. $c = 1/2 - b/3 = 1/4$, $a = 1 - 3/4 - 1/4 = 0$.

$$a = 0, \quad b = \frac{3}{4}, \quad c = \frac{1}{4}.$$

(b) El error de truncamiento para una regla de n puntos exacta hasta grado $\leq n - 1$ tiene la forma $C \cdot f^{(n)}(\xi)/(n!)$ para algún C y $\xi \in (0, h)$. Aquí $n = 3$ (3 puntos: $0, h/3, h$), la fórmula es exacta para grado ≤ 2 (verificar: $f = x^3$ daría error). El término de error es $\frac{C}{3!} f'''(\xi)$; encontramos C :

La regla es exacta para $f = x^2$ (verificado). Para $f = x^3$:

$$\int_0^h x^3 \, dx = \frac{h^4}{4}, \quad h\{b(h/3)^3 + ch^3\} = h\left\{\frac{3}{4} \frac{h^3}{27} + \frac{1}{4} h^3\right\} = h^4 \left(\frac{1}{36} + \frac{1}{4}\right) = \frac{10h^4}{36} = \frac{5h^4}{18}.$$

Error: $\frac{h^4}{4} - \frac{5h^4}{18} = h^4 \left(\frac{9-10}{36}\right) = -\frac{h^4}{36}$. Como $f'''(x) = 6$ para $f = x^3$: error = $\frac{C}{6} \cdot 6 \cdot h^4/\dots$

Formalmente, el error para $f \in C^3[0, h]$ es

$$E = \int_0^h f \, dx - h\{bf(h/3) + cf(h)\} = \frac{C}{3!} f'''(\xi).$$

Para $f = x^3$: $C/6 \cdot 6 = -h^4/36 \Rightarrow C = -h^4/36$. ■

ENUNCIADO — PROBLEMA 3

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SOLUCIÓN — PROBLEMA 3

Problema 3 — Cota del error de interpolación cuadrática de $\sqrt{1+x}$

En los nodos $x_{j-1} = x_j - h$, x_j , $x_{j+1} = x_j + h$ (equiespaciados con paso h):

Por la fórmula del error: $|f(x) - P_2(x)| = \frac{1}{6} |(x - x_{j-1})(x - x_j)(x - x_{j+1})| |f'''(\xi)|$.

Para $f(x) = \sqrt{1+x}$: $f' = \frac{1}{2}(1+x)^{-1/2}$, $f'' = -\frac{1}{4}(1+x)^{-3/2}$, $f''' = \frac{3}{8}(1+x)^{-5/2}$. En $[0, 1]$:
 $|f'''(x)| = \frac{3}{8}(1+x)^{-5/2} \leq \frac{3}{8}$.

Acotando $|(x - x_{j-1})(x - x_j)(x_{j+1} - x)|$ **para** $x \in [x_{j-1}, x_{j+1}]$: Sea $s = x - x_j \in [-h, h]$. El producto es $|(s + h) \cdot s \cdot (s - h)| = |s|(h^2 - s^2)$. Maximizamos $\phi(s) = |s|(h^2 - s^2)$ para $s \in [0, h]$: $\phi'(s) = h^2 - 3s^2 = 0 \Rightarrow s = h/\sqrt{3}$. $\phi(h/\sqrt{3}) = (h/\sqrt{3})(h^2 - h^2/3) = (h/\sqrt{3}) \cdot (2h^2/3) = 2h^3/(3\sqrt{3})$.

Por lo tanto:

$$|f(x) - P_2(x)| \leq \frac{1}{6} \cdot \frac{2h^3}{3\sqrt{3}} \cdot \frac{3}{8} = \frac{h^3}{6} \cdot \frac{2}{3\sqrt{3}} \cdot \frac{3}{8} = \frac{h^3 \cdot 6}{6 \cdot 24\sqrt{3}} = \frac{h^3}{24\sqrt{3}}. \blacksquare$$

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5. Consider the following initial value problem

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where y_0 and s_0 are given constants.

- Write this as a system of first order differential equations.
 - Describe a procedure using the Forward Euler method to approximate the solution to this problem at $t = h$ (i.e. apply one time step of Forward Euler with time step h and find an expression for the approximate solution).
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 - Discuss the general advantages and disadvantages of the Forward and Backward Euler methods for solving an initial value problem.
6. (a) Suppose that A is an $n \times n$ matrix. Show that the spectral radius $\rho(A)$ is always less than or equal to any (induced) matrix norm $\|A\|$.
- (b) Suppose A is a 5×5 matrix with eigenvalues $-5, -4, -1, 0.1, 2$.

Given a random starting vector $\mathbf{x}_0 \in \mathbb{R}^5$, consider the sequence

$$\frac{\mathbf{x}_0}{\|\mathbf{x}_0\|}, \frac{A\mathbf{x}_0}{\|A\mathbf{x}_0\|}, \frac{A^2\mathbf{x}_0}{\|A^2\mathbf{x}_0\|}, \frac{A^3\mathbf{x}_0}{\|A^3\mathbf{x}_0\|}, \dots$$

Show that this sequence converges and identify to what quantity it converges.

7. The QR algorithm for the matrix A is given by

$$A_0 = A = Q_1 R_1 \tag{1}$$

$$A_1 = R_1 Q_1 = Q_2 R_2 \tag{2}$$

$$\dots \tag{3}$$

$$A_k = R_k Q_k$$

- (a) If A is symmetric with eigenvalues $|\lambda_1| > \dots > |\lambda_n|$, how does one find the eigenvalues and eigenvectors of A using QR algorithm?

- (b) Having a QR factorization of the full-rank matrix A ($A = QR$) with $m \geq n$, let $\hat{x}$ be the solution to the least squares problem $\min_x \|b - Ax\|$. Show that $\|b - A\hat{x}\|^2 = \|c_2\|^2$, where c_2 is the vector consisting of the last $m - n$ components of Q^*b .

- (c) Find an orthogonal matrix G and a number w such that

$$G \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} w \\ 0 \end{bmatrix}$$

SOLUCIÓN — PROBLEMA 4

Problema 4 — Singularidad numérica por punto flotante

$\epsilon_{\text{mach}} \approx 10^{-16}$, $\text{UFL} \approx 10^{-308}$, $\text{OFL} \approx 10^{308}$.

Matriz $A = \begin{pmatrix} \epsilon & 0 \\ 0 & \epsilon \end{pmatrix}$: $\det A = \epsilon^2$ (en aritmética exacta). En punto flotante, $\text{fl}(\epsilon \cdot \epsilon) = \epsilon^2(1 + \delta_1)$.

La dificultad ocurre cuando ϵ^2 cae por debajo del nivel de **underflow**: $\epsilon^2 < \text{UFL} \approx 10^{-308}$, es decir $\epsilon < 10^{-154}$. Para $\epsilon \lesssim 10^{-154}$, ϵ^2 se redondea a 0 y A parece singular.

$$\epsilon_A \approx 10^{-154}.$$

Matriz $B = \begin{pmatrix} 1 & 1 + \epsilon \\ 1 - \epsilon & 1 \end{pmatrix}$: $\det B = 1 - (1 + \epsilon)(1 - \epsilon) = 1 - (1 - \epsilon^2) = \epsilon^2$. En punto flotante: $\text{fl}((1 + \epsilon)(1 - \epsilon)) = \text{fl}(1 - \epsilon^2)$. Si $\epsilon^2 < \epsilon_{\text{mach}}$, la operación $1 - \epsilon^2$ pierde toda la información sobre ϵ^2 por **cancelación catastrófica**: $\text{fl}(1 - \epsilon^2) = 1$, y $\det B$ se calcula como 0. Esto ocurre cuando $\epsilon^2 < \epsilon_{\text{mach}} \approx 10^{-16}$, es decir $\epsilon < 10^{-8}$.

$$\epsilon_B \approx 10^{-8}.$$

Diferencia clave: A falla por underflow (ϵ muy pequeño almacena como 0), B falla por cancelación catastrófica en la resta de casi iguales (ocurre mucho antes).

ENUNCIADO — PROBLEMA 5

5. Consider the following initial value problem

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where y_0 and s_0 are given constants.

- Write this as a system of first order differential equations.
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SOLUCIÓN — PROBLEMA 5

Problema 5 — EDO de segundo orden; Euler explícito e implícito

$$y'' = -y - \sin y, y(0) = y_0, y'(0) = s_0.$$

(a) Sistema de primer orden: $u = y, v = y'$:

$$\begin{pmatrix} u \\ v \end{pmatrix}' = \begin{pmatrix} v \\ -u - \sin u \end{pmatrix}, \quad \begin{pmatrix} u(0) \\ v(0) \end{pmatrix} = \begin{pmatrix} y_0 \\ s_0 \end{pmatrix}.$$

(b) Euler explícito un paso (h):

$$\begin{pmatrix} u_1 \\ v_1 \end{pmatrix} = \begin{pmatrix} u_0 + hv_0 \\ v_0 + h(-u_0 - \sin u_0) \end{pmatrix} = \begin{pmatrix} y_0 + hs_0 \\ s_0 - h(y_0 + \sin y_0) \end{pmatrix}.$$

(c) Euler implícito un paso (h):

$$\begin{pmatrix} u_1 \\ v_1 \end{pmatrix} = \begin{pmatrix} u_0 + hv_1 \\ v_0 + h(-u_1 - \sin u_1) \end{pmatrix}.$$

Esto es un sistema no lineal en (u_1, v_1) . De la primera ecuación $v_1 = (u_1 - u_0)/h$. Sustituyendo en la segunda:

$$\frac{u_1 - u_0}{h} = v_0 + h(-u_1 - \sin u_1) \implies \frac{u_1 - u_0}{h} - v_0 + hu_1 + h \sin u_1 = 0.$$

Esta ecuación escalar en u_1 se resuelve **numéricamente** con Newton o bisección (dado que u_1 afecta el término $\sin u_1$). Una vez hallado u_1 , se recupera $v_1 = (u_1 - u_0)/h$.

(d) Ventajas y desventajas:

Euler explícito	Euler implícito	
Implementación	Simple; no se resuelve sistema	Requiere resolver sistema (lineal o no lineal) en cada paso
Estabilidad	Condionalmente estable ($h \leq 2/ \lambda $)	Incondicionalmente estable (A-estable)
Costo por paso	Barato (una evaluación de f)	Caro (iteración de Newton)
Adecuado para	Problemas no rígidos con h pequeño	Problemas rígidos (stiff)

ENUNCIADO — PROBLEMA 6

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SOLUCIÓN — PROBLEMA 6

Problema 6 — Radio espectral $\leq$ norma; convergencia de potencias iteradas

(a) Sea λ valor propio de A con $|\lambda| = \rho(A)$ y v el vector propio ($Av = \lambda v$, $v \neq 0$). Para cualquier norma matricial inducida:

$$\rho(A)\|v\| = |\lambda|\|v\| = \|\lambda v\| = \|Av\| \leq \|A\|\|v\|.$$

Dividiendo entre $\|v\| > 0$: $\rho(A) \leq \|A\|$. ■

(b) A es 5×5 con valores propios $-5, -4, -1, 0.1, 2$. Los vectores propios $\{v_i\}$ forman una base (asumimos A diagonalizable; si no, se usa forma de Jordan). Escribe $x_0 = \sum_i c_i v_i$ (con c_i casi siempre no nulos). Entonces $A^k x_0 = \sum_i c_i \lambda_i^k v_i$. El valor propio dominante es $|\lambda_{\max}| = |-5| = 5$, así

$$\frac{A^k x_0}{\|A^k x_0\|} \approx \frac{c_1 \lambda_1^k v_1 + c_2 \lambda_2^k v_2 + \dots}{|c_1| |\lambda_1|^k \|v_1\| + \dots}.$$

Como $|\lambda_1| = 5 > |\lambda_2| = 4 > \dots$, los términos con $i \geq 2$ decaen relativamente: $(\lambda_i/\lambda_1)^k \rightarrow 0$. Pero $\lambda_1 = -5$: alternancia de signo. Para la secuencia normalizada $A^k x_0 / \|A^k x_0\|$, cuando $k \rightarrow \infty$:

$$\frac{A^k x_0}{\|A^k x_0\|} \rightarrow \pm v_1 \quad (\text{alternando según paridad de } k).$$

La secuencia converge en el sentido de que $|A^k x_0 / \|A^k x_0\| \mp v_1| \rightarrow 0$ para k par o impar respectivamente. Identifica la cantidad: el **vector propio dominante** v_1 correspondiente al valor propio de mayor módulo, $|\lambda_1| = 5$. ■

ENUNCIADO — PROBLEMA 7

5. Consider the following initial value problem

$$\begin{aligned}\frac{d^2 y}{dt^2} &= -y - \sin y, \\ y(t=0) &= y_0, \\ \frac{dy}{dt}(t=0) &= s_0\end{aligned}$$

where y_0 and s_0 are given constants.

- Write this as a system of first order differential equations.
 - Describe a procedure using the Forward Euler method to approximate the solution to this problem at $t = h$ (i.e. apply one time step of Forward Euler with time step h and find an expression for the approximate solution).
 - Describe a procedure using the Backward Euler method to approximate the solution to this problem at $t = h$ (i.e. apply one time step of Backward Euler with time step h and explain how the approximate solution could be obtained numerically).
 - Discuss the general advantages and disadvantages of the Forward and Backward Euler methods for solving an initial value problem.
6. (a) Suppose that A is an $n \times n$ matrix. Show that the spectral radius $\rho(A)$ is always less than or equal to any (induced) matrix norm $\|A\|$.
- (b) Suppose A is a 5×5 matrix with eigenvalues $-5, -4, -1, 0.1, 2$.

Given a random starting vector $\mathbf{x}_0 \in \mathbb{R}^5$, consider the sequence

$$\frac{\mathbf{x}_0}{\|\mathbf{x}_0\|}, \frac{A\mathbf{x}_0}{\|A\mathbf{x}_0\|}, \frac{A^2\mathbf{x}_0}{\|A^2\mathbf{x}_0\|}, \frac{A^3\mathbf{x}_0}{\|A^3\mathbf{x}_0\|}, \dots$$

Show that this sequence converges and identify to what quantity it converges.

7. The QR algorithm for the matrix A is given by

$$A_0 = A = Q_1 R_1 \tag{1}$$

$$A_1 = R_1 Q_1 = Q_2 R_2 \tag{2}$$

$$\dots \tag{3}$$

$$A_k = R_k Q_k$$

- (a) If A is symmetric with eigenvalues $|\lambda_1| > \dots > |\lambda_n|$, how does one find the eigenvalues and eigenvectors of A using QR algorithm?

- (b) Having a QR factorization of the full-rank matrix A ($A = QR$) with $m \geq n$, let $\hat{x}$ be the solution to the least squares problem $\min_x \|b - Ax\|$. Show that $\|b - A\hat{x}\|^2 = \|c_2\|^2$, where c_2 is the vector consisting of the last $m - n$ components of Q^*b .

- (c) Find an orthogonal matrix G and a number w such that

$$G \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} w \\ 0 \end{bmatrix}$$

SOLUCIÓN — PROBLEMA 7

Problema 7 — Algoritmo QR: valores propios, mínimos cuadrados, Givens

(a) El algoritmo QR: $A_0 = A$, $A_k = R_k Q_k$ (con $A_{k-1} = Q_k R_k$). Para A simétrica con $|\lambda_1| > \dots > |\lambda_n|$: las iteradas A_k convergen a la forma diagonal $\text{diag}(\lambda_1, \dots, \lambda_n)$ (bajo condiciones genéricas). Los valores propios se leen en la diagonal de A_k para k grande; los vectores propios se acumulan como columnas del producto $Q_1 Q_2 \dots Q_k$, que converge a la matriz ortogonal de vectores propios.

(b) (Mínimos cuadrados vía QR) Con $A = QR$ (Q ortogonal $m \times m$): $\|b - Ax\|^2 = \|Q^T(b - QRx)\|^2 = \|Q^T b - Rx\|^2$ (norma invariante bajo ortogonal). Particionando $Q^T b = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$ con $c_1 \in \mathbb{R}^n$, $c_2 \in \mathbb{R}^{m-n}$, y $R = \begin{pmatrix} R_1 \\ 0 \end{pmatrix}$: $\|Q^T b - Rx\|^2 = \|c_1 - R_1 x\|^2 + \|c_2\|^2$. Mínimo en $x = \hat{x}$: $R_1 \hat{x} = c_1$ (sustitución regresiva), y $\|b - A\hat{x}\|^2 = \|c_2\|^2$. ■

(c) $G \begin{pmatrix} 3 \\ 4 \end{pmatrix} = \begin{pmatrix} w \\ 0 \end{pmatrix}$. G ortogonal preserva la norma: $w = \sqrt{9 + 16} = 5$. La rotación de Givens: $c = 3/5$, $s = 4/5$,

$$G = \begin{pmatrix} c & s \\ -s & c \end{pmatrix} = \frac{1}{5} \begin{pmatrix} 3 & 4 \\ -4 & 3 \end{pmatrix}, \quad G \begin{pmatrix} 3 \\ 4 \end{pmatrix} = \frac{1}{5} \begin{pmatrix} 9 + 16 \\ -12 + 12 \end{pmatrix} = \begin{pmatrix} 5 \\ 0 \end{pmatrix}. \quad \checkmark$$

$$\boxed{w = 5, \quad G = \frac{1}{5} \begin{pmatrix} 3 & 4 \\ -4 & 3 \end{pmatrix}.}$$